Thesis review response for thesis titled, "Symmetry based analysis of instabilities in a plate with stiffened edge"

Contents

1 Review 1

1st Report

Indian Institute of Technology Kanpur
Evaluation Report of Ph.d Thesis

Sudhe
Chairman, SPGC

CATEGORY (SPGC): II

1. Name of student: DEEPANKAR DAS

2. Roll No. 17205264 **3. Department** Mechanical Engineering

4. General features of thesis

(i) Organization and Presentation:

The organization and presentations are good. A more detailed account is given in the attached PDF document.

(ii) Is the quality of work comparable with that in other universities of repute?

Yes

(iii) Does the thesis embody any new ideas with original thoughts?

Yes

5. Comments (The examiner may give details on additional sheet(s), if required.)

(i) Corrections for punctuation, grammar, spelling, typographical errors or language:

MINOR

(ii) Technical content of the thesis:

See attached document.

(iii) Highlights and strong/weak points in the thesis:

Strong points:
- Theoretical rigour of the analysis
- Robust framework that can be used to analyse a wide variety of problems
- Compared and validated with corresponding full field finite element simulations

Weak Points:
- More connection with actual experiments
- Framework as described not clear if it can be applied on dissipative solids (viscoelasticity, plasticity etc..._)

6. Suggestions (The examiner may give details on additional sheets.)

I suggest the author to discuss in more detail the limitations and above strengths of the methodology in the conclusion section. Also try to make some contact with existing experiments if any or propose such experiments for future work.

7. Specific Recommendation

The Thesis is acceptable and the corrections, modification and improvement suggested by me be Incorporated in the thesis to the satisfaction of the oral board.

8. SPGC Remark

In Partial Fulfillment of the Requirements for the Degree of

Doctor of Philosophy

at the Mechanical Engineering, Indian Institute of Technology, Kanpur

The thesis comprises a total of six chapters, one of which is the introduction and one closure. It presents a theoretical and numerical study of elastic instabilities in a circular von Kármán plates with stiffened boundary. The mismatch in strain triggered by the stiffened boundary serves as the bifurcation parameter of the problem. The study employs reduction methods using the symmetries of the problem at hand allowing for a rigorous theoretical treatment of the problem and its subsequent numerical implementation. The problem is of great interest in several engineering applications including but not limited to composite structures (at various scales) and biological systems. The methodology and findings of this thesis contribute to the understanding and further development of symmetry-driven bifurcation problems in solid mechanics and engineering.

The introduction (Chapter 1) presents a brief state of the art in bifurcation problems with many symmetries and relevant engineering applications.

Chapter 2 introduces some basic mathematical notation and definitions, as well as important assumptions considered throughout the study. In particular, the stiffened edge of the plate is approached by using a special Cosserat rod model. Subsequently, the full set of nonlinear equilibrium equations and potential energy are introduced. The notion of symmetry groups is defined allowing the reduction of the full set of possible bifurcated
solutions. The determination of the local bifurcation branch is discussed using the
1 Lyapunov-Schmidt method (some typos need to be corrected in the original manuscript
for instance the latter names in p.18 for example). The caption of the figures could be more
detailed to allow the reader a better understanding of the what is shown.

Chapter 3 discusses the linear bifurcation analysis of the plate problem. The
deformation is divided to in-plane and out of plane components owing to the thin
dimension of the plate in the 3rd-direction. This allows to define subsequently a bending
and membrane strain energy for the system at hand. The problem assumes overall small
in-plane stretches but large out of plane deflections and rotations, thus providing the
2 nonlinearities necessary for bifurcation of the plate. It is my impression that in Eq. (3.1.8)
the tensor A has not been defined (perhaps this equation should not be there). Once all
energies and constraints have been defined an augmented potential energy is written. The
first variation and resulting equilibrium equations with boundary conditions and
constraints are then obtained. The trivial homogeneous solution is proposed next. This
allows to carry out next a bifurcation analysis around that principal solution. For this the
problem is rewritten using a nonlinear operator and dimensionless form. The nonlinear
operator is then linearized allowing the solution of the bifurcation problem in a semi-
analytical form. A short concluding remark section for the chapter could allow the reader
3 to grasp the main contribution of the chapter.

Chapter 4 discusses the previous problem reduction by taking into account the
inherent symmetries of the material (isotropy) and geometry (circular plate). The chapter
presents in detail one by one the actions of the symmetry in all relevant scalar, vector and
tensor fields of the problem. At the end of section 4.1, it could be very useful to summarize
4 the main outcomes of the long algebra carried out. To perform the local bifurcation
analysis, the determination of the adjoint operator of the main kernel is necessary. The
algebraic manipulations for this are described in section 4.2.0.1. The results are presented
in a direct flow and summary of the final expressions and their influence on the solutions
are not described. It would help the reader to try to explain the influence of the terms on
the bifurcation results. The paragraph after Eq. (4.2.92) needs to be written more clearly.
Subsequently, the analysis provides the expanded solution up to displacements of second
5 order in the curve parameter. It could be useful the equation, figure and citation references
in the pdf text to be in form of hyper-references so that the reader can easily jump to the
equation or figure of interest. Moreover, it could be more efficient for reading to move
parts of the intermediate algebraic steps in an Appendix of the chapter and keep only initial
hypotheses, important intermediate and final results. Also, one would expect that the semi-
analytical results are used to validate the FEM results and not vice versa. Perhaps the FE
code has already been verified and this allows the cross-check of the results and
expressions obtained by the semi-analytical method.

Chapter 5 presents the results obtained by use of the previous semi-analytical and
numerical methods. It is useful to recall in the beginning of this chapter the loading
parameter λ and its physical interpretation. The different bifurcation modes and critical
loads are shown in the document. One could ask if all those bifurcation modes, predicted
by the analysis, are attainable in reality? What is necessary for this to be achieved? The
6 discussion of the figures in this chapter is discussed succinctly and it lacks some in depth
explanations of the results and the potential differences between the model and the FEM
solutions. One question that one can ask is why the authors used an explicit Abaqus
formulation and not an implicit one, given that in most of the cases the response is
supercritical and thus does not even require an arc-length method? Another question
would be how the Poisson ratio of the membrane affects those results. One could for
instance plot the critical bifurcation loads as a function of this material parameter. Other

parametric studies and plots could easily be reproduced from the existing data to reveal the richness of the responses in such systems.

The thesis concludes with a closure chapter summarizing briefly the main findings
of the thesis. It is concluded that in most of the cases, the post-bifurcation response is
supercritical indicating a stable behavior beyond the bifurcation point. The author states
some limitations of the present analysis. It would have been interesting to comment
further whether the proposed workflow can be potentially extended to take into such large
edge deformations. The future work section mainly raises the applicability of the von
Kármán plate assumptions and the coupling with the edge rod. I would have been
7 interested to know, in addition to that, if and how these approaches can deal with
responses beyond elasticity (such as plasticity, nonlinear elasticity of the membrane,
viscoelasticity etc.). Also do the authors believe that the commercially available codes are
able to solve these problems? What are the precautions needed in such cases of high
symmetry. Finally, in real applications small asymmetries are inevitable due to
manufacturing reasons. Would in that case of minor symmetry breaking a number of
bifurcation modes entirely disappear and some basic ones remain? Which would those be?

As a conclusion, the present thesis presents a theoretical and numerical study of the (post-)bifurcation of a thin plate constrained by an inextensible and shear-free ring at the boundary. The semi-analytical solution of the problem is extremely tedious algebraically and the author manages to get to the bottom of it, which is fairy remarkable. These solutions are validated and further compared with finite element results.

By consideration of the above comments, I am favorable for the defense of this thesis as a fulfillment of the requirements for a Doctorate degree.

1.1 Response:

Thank you for your valuable review and suggestions. I have added more references to the introduction, citing experimental works that study structures similar to the one presented in the thesis. I have also made other modifications to the thesis based on your comments, as follows,

1 I have corrected the typos in the manuscript and expanded the figure captions to be more informative.

2 The tensor $\mathbf{A}$ is used as a place-holder tensor used to show the expression of the function $W(\cdot)$. I have added additional text in Eq. (3.1.8) to avoid the ambiguity.

3 Thank you for your valuable suggestion. I have added a concluding remark at the end of chapter 3 highlighting the outcome of the chapter.

4 Thank you for the suggestion, I have added text at the end of section 4.1 (on page 91) summarizing the outcome of the algebraic manipulations.

5 I have rewritten the paragraph after equation 4.2.92 (4.2.93 in revised thesis, page 117). I have hyper-linked the equations, figures and citations. I have moved some of the intermediate steps to appendix to improve readability. The symmetry FEM algorithm used is an implementation of an well established work on shell buckling, hence I used it to validate the semi-analytical results.

6 Thank you for the detailed feedback and questions. Regarding the physical feasibility of the bifurcation modes, it would require stability analysis of the buckled modes. I have not performed stability analysis, so it is difficult to comment on it authoritatively, but it can be expected that the supercritical modes with lower periodicity should be achievable physically. ABAQUS/Explicit was used for determining the critical points because the inextensibility and unshearability constraints on the plate edge and the hinge-like coupling was causing convergence issues in the implicit solver. Explicit solver didn't face the same issues but had less accurate results. But as the models were different anyway, it was only used to compare the semi analytical results qualitatively. Regarding the parametric studies, I have captioned the parameter sweeps to be more descriptive (page 144).

7 Thank you for the important question. The workflow carried out in the thesis will still hold for large deformations and non linear elasticity, but the flow won't extend to plastic and viscoelastic systems. This is because the core of the work, i.e., Lyapunov Schmidt Reduction depends on the ellipticity of the governing PDEs, which is not the case in plastic and viscoelastic systems. Commercial packages don't have a symmetry detection built into the post-buckling analysis, so they struggle with tracing the post-buckling solution when the solution curves are close together. The symmetry reduced analysis allows us to separate solutions based on the symmetries of the linear buckling analysis. In the current work, we partition the solution space based on the symmetry sub-groups of the whole system and then look for post-buckling behaviour. In presence of non-symmetric defects the current analysis can't detect the solutions arising due to those defects, unless they exist in any of the symmetry group of the problem.

2 Review 2

2nd Report

Indian Institute of Technology Kanpur
Evaluation Report of Ph.d Thesis

Sudhe
Chairman, SPGC

CATEGORY (SPGC): II

1. Name of student: DEEPANKAR DAS

2. Roll No. 17205264 **3. Department** Mechanical Engineering

4. General features of thesis

(i) Organization and Presentation:

See attached

(ii) Is the quality of work comparable with that in other universities of repute?

Yes

(iii) Does the thesis embody any new ideas with original thoughts?

Yes

5. Comments (The examiner may give details on additional sheet(s), if required.)

(i) Corrections for punctuation, grammar, spelling, typographical errors or language:

MINOR

(ii) Technical content of the thesis:

See attached

(iii) Highlights and strong/weak points in the thesis:

See attached

6. Suggestions (The examiner may give details on additional sheets.)

See attached

7. Specific Recommendation

The Thesis is acceptable and the corrections, modification and improvement suggested by me be Incorporated in the thesis to the satisfaction of the oral board.

8. SPGC Remark

Report on: ***Symmetry Based Analysis of Instabilities in a Plate with Stiffened Edge,*** **PhD Dissertation by Deepankar Das.** February 18, 2025

This is a detailed study of an attractive and difficult problem of nonlinear solid mechanics: A flexible thin rod-ring is perfectly attached to a thin circular plate along its boundary. The length of the former differs from the circumference of the plate. This initial incompatibility drives the homogeneous structure to lose stability in favor of other finitely deformed configurations. A natural tool for the nonlinear analysis of such a problem is "bifurcation theory in the presence of a symmetry group". The ratio of the elastic stiffness of the plate to that of the ring is chosen to be the bifurcation parameter in this work. Although well known, the symmetry/bifurcation approach is only rarely employed in the field of nonlinear structural mechanics. Indeed, commercially available finite-element codes without nonlinear-bifurcation capabilities abound, which explains the popularity of ad-hoc, purely computational approaches. In any case, it is both encouraging and refreshing to see this systematic approach undertaken in the dissertation under review.

After a nice introductory chapter on instability and buckling of elastic systems, the plate-ring model is presented along with a summary of the overall symmetry-based approach in Chapter 2. The nonlinear equilibrium equations are derived in Chapter 3. A flat, homogeneous family of solutions is noted, about which the problem is subsequently linearized. In Chapter 4, the equivariant symmetries of the governing equations, viz., appropriate representations of $\mathrm{O}(2)$ and $\mathrm{SO}(2)\times \mathrm{Z}_2$, are exhaustively presented, after which symmetry-reduction methods are employed to obtain local solutions of the nonlinear problem. This is the heart of the dissertation. Many beautiful nontrivial configurations are obtained, and the analytical work is supplemented by a Fourier-finite-element analysis.

Overall, I am impressed by the work and results presented. In particular, the symmetry arguments are correctly implemented yielding some really nice results. There are a couple of technical issues regarding mathematical rigor (from the analysis point of view), discussed below. However, this is a PhD thesis for mechanical engineering – not mathematics. Overall, I think the work represents a huge improvement on the usual guesswork (via imperfections) often used with commercially available software in many engineering studies on instability/bifurcation problems.

Corrections, observations and questions:

1 p. 8, middle of the page: $\mathbf{Ra}\cdot\mathbf{Rb}=\mathbf{a}\cdot\mathbf{b}$

2 p. 13, l. 2: ... of the rod using ...

3 p. 14, Remark 2.2: ... hypothesis (see Assumption 3, p. 21) ...

1st l. after Remark 2.2: ... above-mentioned homogeneous ...

4 p. 17, Theorem 2.4: The *local* bifurcation theorem is valid provided that the operator $\mathcal{L}(\lambda_o):\mathcal{U}\to\mathcal{Z}$ is Fredholm of index zero, i.e., $\dim N(\mathcal{L}(\lambda_o)) = co\dim R(\mathcal{L}(\lambda_o)) < \infty$. This is fulfilled automatically in reference [21], due to the assumed form of the nonlinear operator (the linearization has the form "identity minus a compact linear operator"). Moreover, condition (2) is not generally sufficient when $\dim N(\mathcal{L}(\lambda_o)) > 1$. In that case, one needs the so-called "crossing number" – see Kielhöfer's book. (This

point is missed in [21]). Of course, only the case of a 1D null space is needed throughout the dissertation, and condition (2) is fine in that case.

5 p. 18, l. 1: ... bifurcation analysis has two ...

6 p. 34, l. 1: As the equilibrium ... (or Because the equilibrium ...)

7 p. 35, eq. 3.3.6 and p. 37, eq 3.3.13: Spaces of continuously differentiable functions (with maximum norms so they are Banach) do not yield the Fredholm property in linear elliptic PDE problems. The simplest counterexample is: $\Delta u = f$ in Ω, $u = 0$ on $\partial\Omega$, with $f \in C(\bar{\Omega})$. Then $u \notin C^2(\bar{\Omega})$ in general. (E.g., see "PDEs" by V.P Mikailov, MIR 1978, pp. 246-247.) One has 2 choices to do things rigorously: Either work in Hölder spaces, e.g., $f \in C^{\alpha}(\bar{\Omega}) \Rightarrow u \in C^{2,\alpha}(\bar{\Omega})$, $0 < \alpha \leq 1$, or Sobolev spaces, $f \in L^p(\Omega) \Rightarrow u \in W^{2,p}(\Omega)$, $2 \leq p < \infty$, As it stands, the Liapunov-Schmidt reduction in the dissertation is formal (i.e., algebraically correct but not rigorous).

8 p. 41: Again, for rigorous results, the nonlinear mapping should be Fréchet differentiable. Differentiability for the problem treated in the dissertation should be fine – the equations are semi-linear. See Kielhöfer's book for this and for a discussion on the function spaces mentioned above.

9 p. 49: The mechanical system studied here is apparently unrestrained, i.e., free boundaries(?). Thus, I would expect to see a 6-dimensional "rigid" null space for all parameter values. On the other hand, I suspect that the fixed-point spaces (after symmetry reduction) eliminate these(?)

10 p. 63: Remark: It is typically much easier to first establish the *invariance* of the total potential energy under the group action. Equivariance then follows directly by taking the first variation of the invariance relations.

11 pp. 125-128: Beautiful results!

p. 130, l. 6: This is often possible by fine tuning parameters – but it's non-generic. This is usually referred
12 to as "accidental degeneracy" in quantum mechanics.

Par. 2: How was ABAQUS employed? As mentioned before - what about rigid modes? Also, ABAQUS has neither bifurcation nor stability capabilities. Thus, imperfections must have been employed(?) Speaking of *stability*, the connection between stability/instability and super-critical/subcritical bifurcation should be mentioned early on, say, in Chapter 2.

13 pp. 133-135: Is there a discussion about these figures – in particular, Figure 5.9?

14 p. 142, par. 2: The von-Kármán model entails "moderate rotations + small strains". Large stretching requires more, e.g., see [37].

2.1 Response:

Thank you for your insightful review and suggestions. I have made modifications to the thesis based on your comments as follows,

1-3, 5-6 Thank you for pointing out the errors. I have corrected the mentioned typos.

4 Thank you for the insightful comment. I have added the restrictions to the theorem 2.4 on page 18.

7-8 Thank you for pointing out the issue on mathematical rigour. I had overlooked it during the analysis. I have redefined the spaces to Hölder spaces with sufficient differentiability on page 38.

9 You are correct with the comment on the rigid null space, except that rigid modes are four dimensional as rotation about diameter is not degenerate in von-Kármán model. In the thesis I had manually omitted the rigid mode coefficient, i.e., modes corresponding to B_0, A_1, B_1 and E_0. I have summarised this omission as remark 3.20 at the end of chapter 3 on page 60.

10-11 Thank you for the valuable feedback and insight.

12-13 I have added more details about the ABAQUS model in the thesis (page 145-146) clarifying the usage ans explaining the figures. It is noted that ABAQUS is only used to determine critical values for preliminary validation. The numerical post-buckling is carried out using symmetry reduced finite element method using MATLAB codes based on wohlever's work.

14 Thank you for pointing it out, I have rectified the text in the thesis.

3 Review 3

3rd Report

Indian Institute of Technology Kanpur

Evaluation Report of Ph.d Thesis

Sudhe
Chairman, SPGC

CATEGORY (SPGC): II

1. Name of student: DEEPANKAR DAS

2. Roll No. 17205264 **3. Department** Mechanical Engineering

4. General features of thesis

(i) Organization and Presentation:

The thesis report is more or less written in a readable form. Overall the huge effort of the student is clearly visible. Suggestion: Introduction and Conclusion need to be revised to address important points observed in the analysis and their implications on a bigger class of problems.

(ii) Is the quality of work comparable with that in other universities of repute?

Yes

(iii) Does the thesis embody any new ideas with original thoughts?

Yes

5. Comments (The examiner may give details on additional sheet(s), if required.)

(i) Corrections for punctuation, grammar, spelling, typographical errors or language:

MINOR

(ii) Technical content of the thesis:

The problem tackled is new and difficult in the existing set of problems analyzed in theoretical framework of solid mechanics and structural mechanics.
The technique of local bifurcation analysis that has been employed is sound, pertinent and rigorous relative to an engineering department. Numerical methods supplementing the analytical and semi-analytical results add to the richness of the analysis and results. The student deserves the credits for putting such a strenuous effort into understanding the vocabulary, definitions, theorems and their implementation in analysis of a seemingly monstrous problem.

(iii) Highlights and strong/weak points in the thesis:

Strength of the thesis lies in an exhaustive nature of analysis for local bifurcation analysis with symmetry along with implementation of a numerical (bifurcation/continuation) algorithm to solve the same problem. Weak points are specially in introduction and conclusion where more efforts need to be put to explain the main contributions of thesis for a non-technical reader in engineering (as suggestion in the item (i)), that is both chapter 1 and chapter 6 may be expanded further.

6. Suggestions (The examiner may give details on additional sheets.)

7. Specific Recommendation

The Thesis is acceptable and the corrections, modification and improvement suggested by me be Incorporated in the thesis to the satisfaction of the oral board.

8. SPGC Remark

3.1 Response:

Thank you for your valuable review and suggestions. I have made modifications to the thesis based on your comments. I have added more references to expand on engineering applications in Introduction (chapter 1) and added a paragraph regarding applicability of the analysis in Conclusion (chapter 6) with more focus on engineering applications.

4 Review 4

4th Report

Indian Institute of Technology Kanpur
Evaluation Report of Ph.d Thesis

Sudhe
Chairman, SPGC

1. Name of student: DEEPANKAR DAS

CATEGORY (SPGC): II

2. Roll No. 17205264

3. Department Mechanical Engineering

4. General features of thesis

(i) Organization and Presentation:

The author has considered a problem of bifurcations of a circular thin plate bounded by a stiff edge under strain incompatibility between the plate and bounding edge. After introducing the Von-Karman model for the plate and the Cosserat rod model for the edge, the equilibrium equations are derived for the plate-rod system. The perturbed governing equations obtained after linearization then yield the null-space for the in-plane and out-of-plane deformations of the plate-rod system. The author explains the symmetries of the trivial state. Using a standard Lyapunov-Schmidt reduction, the local bifurcation analysis is presented. The bifurcation diagrams obtained analytical and numerically using FEM are discussed at the end of thesis.
The thesis is detailed in its approach. The thin plates bounded by a stiff rod is a simplified yet useful model to study a variety of natural and engineering systems. The use of group theoretic approach to a bifurcation for these systems is of interest.

(ii) Is the quality of work comparable with that in other universities of repute?

Yes

(iii) Does the thesis embody any new ideas with original thoughts?

Yes

5. Comments (The examiner may give details on additional sheet(s), if required.)

(i) Corrections for punctuation, grammar, spelling, typographical errors or language:

MINOR

(ii) Technical content of the thesis:

(iii) Highlights and strong/weak points in the thesis:

6. Suggestions (The examiner may give details on additional sheets.)

My comments and suggestions are given in the attached PDF file.

7. Specific Recommendation

The Thesis is acceptable and the corrections, modification and improvement suggested by me be Incorporated in the thesis to the satisfaction of the oral board.

8. SPGC Remark

Review for the thesis titled: "Symmetry based analysis of instabilities in a plate with stiffened edge" by Deepankar Das

The author has considered a problem of bifurcations of a circular thin plate bounded by a stiff edge under strain incompatibility between the plate and bounding edge. After introducing the Von-Karman model for the plate and the Cosserat rod model for the edge, the equilibrium equations are derived for the plate-rod system. The system of equations is linearized about the base state of uniform stretching/compression. The perturbed governing equations then yield the null-space for the in-plane and out-of-plane deformations of the plate-rod system. The author explains the symmetries of the trivial state and goes on to derive the equivariance of the governing equations. Using a standard Lyapunov-Schmidt reduction, the local bifurcation analysis is presented. The bifurcation diagrams obtained analytical and numerically using FEM are discussed at the end of thesis.

The thesis is detailed in its approach. The thin plates bounded by a stiff rod is a simplified yet useful model to study a variety of natural and engineering systems. The use of group theoretic approach to a bifurcation of thin plates under the edge constraint is of interest.

A few minor comments and questions are given below.

1. Pg. 10: The configuration with $\widehat{\Omega}, \widehat{\Gamma}$ shown in Figure 2.1 is not introduced in the text below.
2. Pg. 12: How is $\left\{\widetilde{d_1}^{\dagger}, \widetilde{d_2}^{\dagger}, \widetilde{d_3}^{\dagger}\right\}$ related to the standard basis $\{e_1, e_2, e_3\}$?
3. Pg. 14: The Assumption 3 in Remark 2.2 appears latter in the text.
4. Pg. 21: The symbol 'Sym' is not used in the text. Does it refer to the Eq. 3.1.1.
5. Pg. 22: The first constraint given in Eq. 3.1.5 corresponds to the in-extensibility of the rod. The text above Eq. 3.1.5 says otherwise.
6. Pg. 24: Eq. (3.2.2) is introduced as a constraint. The author can explain why this condition is needed as an independent constraint in addition to Eq. (3.1.6). In fact, the Lagrange multiplier (β) associated to the constraint goes to zero, as shown by the author on Pg. 31.
7. Pg. 24: Again in Eq. (3.2.2), the symbol for Lagrange multiplier and the bending modulus in Eq. (3.1.9) is the same, which should be rectified.
8. Pg. 34, paragraph 3: The vectors e_ϑ and e_r are not defined in the text.
9. Pg. 35: The symmetry of $\mathbf{P}$ results from the symmetry of $\mathbf{N}$.
10. Pg. 41: Define the variables $\boldsymbol{s}$ and $\boldsymbol{n}$ in Eq. (3.4.1).
11. Pg. 41: Provide a derivation or a reference for Eq. (3.4.4).
12. Pg. 43: Define $\boldsymbol{X}_\lambda$ in Eq. (3.4.12).
13. Pg. 46: Define the differential operator D_ϑ in Eq. (3.4.21)
14. Pg. 48: The observation that the range of k being finite is mentioned in the paragraph at the end of Pg. 48 and at the beginning of Pg. 49. It is suggested that author provide a clearer explanation of why is this the case.
15. Pg. 57: An illustration, may be through a schematic, of the different symmetry operations discussed at the end of Pg. 57 would be helpful.
16. Pg. 58: Is ef an element of G in Eq. (4.1.1)?

17. Pg. 58: After Eq. (4.1.2), it is mentioned that the "null vector is static under action of the elements..". What does the author mean by this statement?
18. Pg. 59: Is $\mathbf{T}_\phi$ the representation of SO(2) on the configuration space $\mathbf{x}$?
19. Pg. 65: It is suggested to write a couple more steps to the derivation of Eq. (4.1.47). Similarly, on Pg. 73 for Eq. (4.1.94).
20. Pg. 81: Is $\boldsymbol{V}$ the invariant subspace of T_{D_n} in Eq. (4.1.138)?
21. Pg. 103: What is the expression for the adjoint in-plane null vector Z_k?
22. Pg. 107: Is the second derivative $\ddot{\lambda}(0)$ obtained numerically in Eq. (4.2.79b)?
23. Pg. 123: Provide a description of the FEM discretized version of the governing equations.

(i) Describe the degrees of freedom for the rod and the plate and their continuity at the interface.

(ii) How was the primary path obtained using the continuation algorithm?

24. Pg. 125: Are the bifurcation diagrams shown in Fig. 5.1 obtained semi-analytically?
25. Pg. 140: In Figure 5.16, the bifurcation diagram obtained using FEM diverges from the theoretically calculated curve. If the curve is continued further, does it encounter a limit point in λ? What does the global bifurcation diagram looks like? It is suggested that the author plot a few configurations along this curve.

4.1 Response:

Thank you for your valuable review and suggestions. I have made modifications to the thesis based on your comments as follows,

1-5, 7-13 Thank you for the detailed feedback and pointing out the errors. I have rectified the errors and provided relevant definitions wherever required.

6 The additional constraint is needed as the constraint equation is effectively a two dimensional vector in three dimensional space. Therefore an additional constraint was needed to account for the third dimension of the Lagrange multiplier. I have added text on page 26 and page 33 of the thesis to emphasize the same.

14 Thank you for feedback, I have added additional text to explain the finite range of k corresponding to the planar critical values on page 53.

15 Thank you for the feedback, I have added a schematic in the thesis, demonstrating the operations in figure 4.1 on page 64.

16 Yes, ef is an element of $\mathcal{G}$, in fact $ef = fe = fer_{2\pi}$.

17 I have added additional text to clarify the statement "null vector is static under action of the elements...". In brief, it means that the null solution doesn't change under application of operations corresponding to said elements of the group.

18 Yes it is the rotation tensor corresponding to elements of $SO(2)$.

19 Thank you for the feedback, I have added the additional derivation steps on page 71 and page 79.

20 Yes $\boldsymbol{V}$ is an invariant subspace of $\mathcal{T}_{D_n}$, in particular it is the eigen subspace of $\mathcal{T}_{D_n}$ corresponding to unity eigen value.

21 Thank you, I have moved the expression of adjoint null vector Z_k from the appendix to main text to the main text.

22 Yes, $\ddot{\lambda}(0)$ is obtained semi analytically, i.e., the critical value λ is obtained numerically, and then corresponding $\ddot{\lambda}(0)$ is obtained by substituting λ in symbolic expressions.

23 Thank you for the valuable feedback. I have added a brief description of the discretization and primary path solution process as a separate chapter after chapter 4.

24 Yes the bifurcation diagrams are obtained semi analytically.

25 Thank you for raising an important point. I tried to compute the curve further but faced convergence issue close to the limit point. It might be due to the von-Kármán model being unable to model large rotations where sections of the rod rotate sharply.